RE PREDICT

AI Powered Renewable Energy Intelligence and Forecasting Platform

Zinnovatio 4.0 Hackathon Project Report

Sense -> Understand -> Predict -> Detect -> Simulate -> Explain

Prepared for hackathon judges, mentors, and implementation teams
2 October 2026

Data foundation: Central Electricity Authority Renewable Energy Portal

The Central Electricity Authority (CEA) remains the authoritative source for its published data. RE-PREDICT is a proposed analytical and decision-support layer; it does not replace official reporting.

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1 Executive Summary

RE-PREDICT is a proposed AI-powered renewable-energy intelligence platform built on the official Central Electricity Authority Renewable Energy Portal. Rather than duplicating a report repository, it turns official published generation and capacity reporting into a decision-support workflow: users can trace what happened, inspect trends, forecast a defined target, flag unusual behavior, test clearly labeled what-if capacity scenarios, and obtain evidence-grounded explanations.

CEA states that its dedicated Renewable Energy Portal publishes daily, monthly, and quarterly renewable-generation reports. The reporting is source-wise for solar, wind, hydro, geothermal, biomass, and bagasse, and includes under-construction project reporting. At the time of this report, the reports page lists daily reports through 30 September 2026. The portal dashboard exposes CO2 emissions mitigated, daily generation for wind, solar and other sources, installed capacity, daily and monthly trend views, regional installed-capacity views, and historical highs. [1][2]

The hackathon MVP should prove a narrow but credible vertical slice: ingest selected official reports, preserve their provenance, normalize only observed fields, provide an exploration dashboard, generate a backtested baseline forecast, show a residual-based anomaly, and execute a transparent scenario calculation. No model scores, financial values, emissions reductions, or real-world impacts are asserted before measurement.

Decision	Recommendation
Positioning	An intelligence layer on official reporting, not a replacement for CEA.
MVP focus	Daily national/source-wise generation analysis with provenance and backtesting.
Trust principle	Every displayed value links to source report, extraction status, and version.
Success proof	Measured data coverage, reproducible evaluation, usable demo flow, and transparent caveats.

2 Project Identity Tagline and Pitch

Project name: RE-PREDICT. Tagline: Renewable intelligence that sees the next signal. One-line pitch: RE-PREDICT layers explainable forecasting, anomaly detection, and scenario analysis on authoritative renewable-energy reporting so users can understand what happened and responsibly explore what may happen next.

3 Zinnovatio 4.0 Track Alignment

The concept aligns with an innovation track centered on AI, sustainability, data engineering, and decision support. It translates routinely published public-sector energy information into an auditable product workflow. The innovation is not claiming a new official dataset; it is the governed path from report ingestion to evidence-linked insight, prediction, detection, simulation, and explanation.

4 Problem Landscape and Statement

Renewable generation is variable, reporting is periodic, and decision makers often need a timely interpreted view. A report archive is essential evidence, but users may still need cross-period comparison, consistent normalized views, forecast uncertainty, anomaly triage, and scenario exploration. Problem statement: How might we convert official renewable reporting into a trustworthy intelligence layer that preserves source context while helping different users reason about past, near-term, and hypothetical renewable outcomes?

5 Product Gap

CEA reporting is the authoritative record and already provides valuable reports and dashboard views. RE-PREDICT complements that foundation: it proposes machine-readable normalization, version-aware trend analysis, backtested forecasts, anomaly explanations, and explicitly hypothetical simulations. It must never imply that a derived estimate, alert, or score is CEA-certified.

6 Vision Mission Objectives Scope and Non Goals

Area	Definition
Vision	Make official renewable reporting easier to interrogate and responsibly operationalize.
Mission	Create a provenance-first intelligence layer for renewable-generation analysis.
Objectives	Ingest reliably; expose trends; forecast with uncertainty; flag anomalies; simulate assumptions; explain outputs.
In scope	Published CEA RE reports and portal-derived context; proposed extensions with governed connectors.
Out of scope	Replacing CEA publication; dispatch control; regulatory advice; unsupported project-level claims; autonomous operational decisions.

7 Target Users Personas and Use Cases

Persona	Need	Representative use case
Energy analyst	Fast, traceable comparison	Compare solar and wind daily generation against prior periods; open underlying report.
Planning or policy team	Structured outlook	Review a 7-day forecast and uncertainty band before a planning discussion.
Renewable developer	Contextual market intelligence	Explore aggregate source trends; never infer confidential plant performance.
Researcher or student	Reproducible evidence	Download a versioned normalized series and methodology notes.
Operations viewer	Early attention signal	Receive an anomaly alert with baseline, observed value, source status, and limitations.

8 Product Concept and End to End User Journey

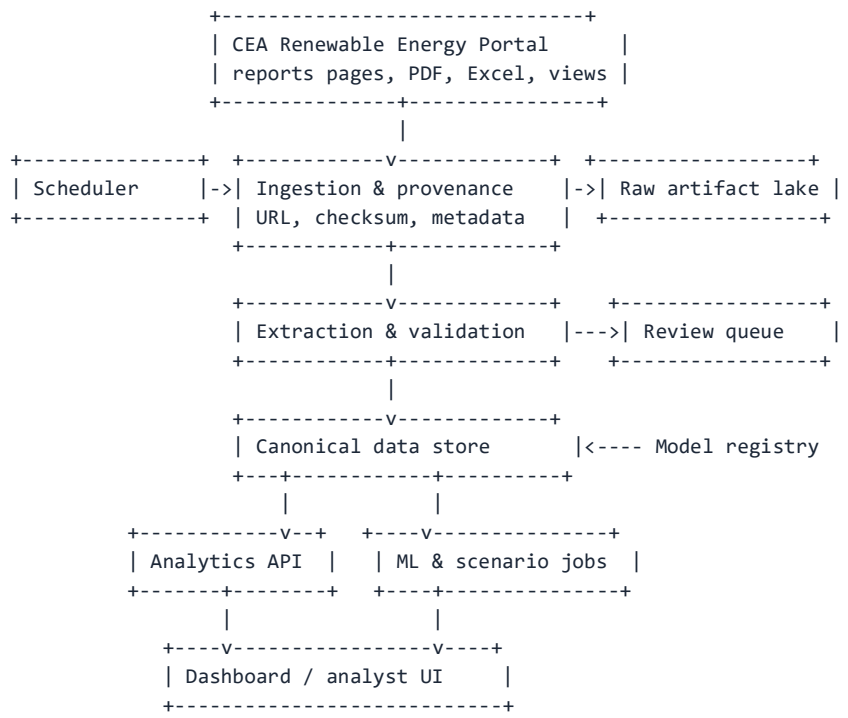
```
Official report -> Ingest & preserve -> Extract & validate -> Normalize -> Analyze
      -> Forecast / detect / simulate -> Explain -> User decision with source link
```

A user selects a period, source and geography available in the normalized data. They see observed values, completeness status, and source links; choose a forecast horizon; review an interval and the backtest card; open an alert only as a hypothesis; change an explicit scenario assumption; then export a cited analysis note. The product labels observed, derived, predicted, and simulated values distinctly.

9 Core Feature Set

Feature	MVP behavior	Trust boundary
CEA ingestion	Catalog report URL, date, format, checksum and retrieval time.	Never alter source artifact.
Analytics	Trends, comparisons, aggregation only where compatible.	Show units and coverage.
Forecasting	Backtested source-wise target and interval.	Label model/version and horizon.
Anomaly detection	Residual and robust deviation flags.	Alert is not root cause.
Explainability	Top drivers, comparisons and source citations.	No causal claim without evidence.
Scenario simulator	Capacity/growth assumption inputs and outputs.	Simulation, not prediction.
Carbon estimate	Transparent factor-based calculation.	Factor is user/version selected.
Reliability score	Team-created analytical score.	Not a CEA metric.
Analyst interface	Natural-language queries over approved semantic layer.	Citations, constrained answers, refusal on unsupported facts.
Alerts	Threshold or model-based notifications.	Human review required.

10 Detailed System Architecture



Component	Responsibility
Ingestion	Discover permitted official artifacts; record retrieval metadata and immutable raw copy.
Extraction	Parse layout-aware PDF/Excel/report page into candidate fields; retain confidence and cell/page locator.
Validation	Schema, unit, date, duplicate, range, reconciliation and freshness checks.
Canonical store	Serve approved observations and provenance IDs.
ML services	Train, evaluate, register, infer and monitor only approved feature sets.
API and UI	Enforce access, expose semantic queries, trace every visual to method and source.

11 Data Architecture

11.1 Source Catalog and Ingestion

The initial catalog contains the CEA Renewable Energy Portal and CEA reports page [1][2]. CEA describes daily, monthly and quarterly reporting; the quarterly material includes under-construction RE project reporting. Each ingest records source URL, report title/date, publication/retrieval timestamp where observable, content type, checksum, parser version, extraction status, and a provenance identifier. Do not predefine report columns: the parser maps only fields actually present in a reviewed source artifact.

Layer	Proposed contents	Quality control
Raw	Original file/page, checksum, retrieval metadata	Immutable object and duplicate checksum detection
Extracted	Candidate labels/values, page/table/cell locator, parser confidence	Human review queue for ambiguity
Canonical	Observation, unit, source type, period, geography as available	Schema, unit, date and reconciliation checks
Serving	Aggregates, features, forecasts, alerts, scenarios	Version joins to raw and canonical records

11.2 Proposed Canonical Schema

Entity	Key proposed fields
source_document	document_id, source_url, title, report_date, retrieved_at, checksum, format, parser_version
observation	observation_id, document_id, period_start, period_end, frequency, source_type, geography, measure_name, value, unit, quality_status
capacity_observation	capacity_id, document_id, period, source_type, geography, capacity_value, unit
extraction_evidence	evidence_id, observation_id, locator, raw_label, raw_value, confidence, reviewer_status
forecast_run	run_id, target_definition, data_version, model_version, horizon, created_at
forecast_point	run_id, timestamp, point_estimate, lower_bound, upper_bound
scenario_run	scenario_id, assumptions_json, formula_version, created_at
alert	alert_id, observation_id, method, severity, status, evidence_json

Data-quality checks: required identifiers; valid period ordering; unit compatibility; non-negative checks only where conceptually valid; duplicate keys; unexpected frequency changes; source-to-extract reconciliation; parser confidence thresholds; and manual approval for material ambiguities. Versioned transformations and source checksums make every output reproducible.

12 ML Architecture

Design decision	Proposed approach
Forecast targets	Observed generation by source, geography and frequency only when sufficient history and stable definition exist.
Horizons	Nowcast/short-term and weekly/monthly horizons are separately evaluated; no universal claim.
Features	Lags, rolling statistics, calendar variables, source/category, capacity where compatible, and later governed weather inputs.
Baselines	Seasonal naive and classical time-series model before gradient boosting.
Candidate models	LightGBM or XGBoost for tabular lag features; classical ARIMA/ETS baseline; LSTM/TFT only later if data breadth justifies it.
Split	Time-ordered train/validation/test and walk-forward validation.
Leakage prevention	Fit transforms on train only; lag features only use information available at forecast issue time; no future revised values.
Selection	Choose by predeclared primary error, interval behavior, stability, latency, and explainability.

Training pipeline: approved canonical snapshot -> feature build with as-of timestamp -> candidate training -> walk-forward evaluation -> experiment log -> review gate -> registry -> batch/API inference. A model card describes target, data coverage, exclusions, feature availability, validation windows, metrics, intended use and limitations.

13 Forecast Evaluation

All criteria below are targets to be measured, not reported results. Evaluation compares like-for-like periods and reports sample size, coverage, target definition and model version.

Metric	Definition	Target success criterion to measure
MAE	Mean absolute error: average |actual - forecast|.	Improve versus seasonal-naive baseline on held-out windows.
RMSE	Square root of mean squared error; weights large misses more.	No degradation versus baseline beyond pre-set tolerance.
sMAPE	Average symmetric percentage error; guard against zero denominators.	Use only when denominator behavior is acceptable.
MASE	MAE scaled by in-sample naive error.	Below 1 suggests better than naive comparator.
R2	Explained variance relative to mean baseline.	Use only with appropriate continuous target and disclose caveats.
Interval coverage	Share of actuals within nominal prediction interval.	Coverage close to stated nominal level.

Metric	Definition	Target success criterion to measure
Calibration	Agreement of forecast probability/interval confidence with outcomes.	Reliability plot and interval-width trade-off reviewed.

14 Anomaly Detection Design

The first detector combines forecast residuals with rolling robust statistics. Let r_t = y_t - yhat_t. A robust z score may be computed as z_t = 0.6745(r_t - median(r))/MAD(r), with safeguards when MAD is zero. Isolation Forest may be evaluated on engineered residual, lag, and seasonal features as a secondary detector. Severity is a proposed policy based on magnitude, persistence, data quality, and operational relevance.

Severity	Proposed rule	Response
Info	Small deviation or data-quality uncertainty	Log; no escalation.
Watch	Material robust deviation or repeated residual	Show explanation and source status.
Review	Large/persistent deviation with valid data	Route to human analyst.
Critical	Only pre-approved, context-specific threshold	Notify named owner; never infer cause automatically.

Root-cause analysis is limited: a residual can identify that behavior differs from baseline, not why. The product can surface contemporaneous report changes, missingness, trend context, and later approved weather signals; it cannot diagnose curtailment, equipment failure, policy action, or grid events without corroborating data.

15 Scenario Engine

A scenario is a transparent conditional calculation, not a prediction. For source s and period t, proposed generation is G_s,t^scenario = C_s,t^scenario x CF_s,t^assumption x H_t, where C is capacity, CF is an assumed capacity factor, and H is hours in period. Alternatively a baseline generation series can be scaled by a documented capacity ratio, subject to the explicit assumption that utilization holds. Constraints include non-negative capacity, valid units, compatible frequency, and ranges selected by the user or scenario template.

Scenario	Input assumptions	Output and uncertainty
Base	Observed/selected capacity and historical reference rule	Reference trajectory, not a forecast.
Growth	Capacity increment/timing and utilization assumption	Range under low/base/high assumption sets.
Mix shift	Source allocation and commissioning timing	Source-wise and aggregate conditional outcome.
Stress	Reduced utilization or availability assumption	Sensitivity view; no causal probability claimed.

16 Carbon Impact Module

Carbon-impact estimates must be factor-based and transparent. Proposed avoided emissions: AE = G x EF_counterfactual, where G is renewable generation in MWh and EF is a documented counterfactual emissions factor in tCO2e/MWh. If a lifecycle renewable factor is also included, report net estimate = G x (EF_counterfactual - EF_RE). The UI must display factor source, geography/time applicability, version, units, uncertainty range, and that this is an estimate. It must not present an unsupported aggregate emissions reduction as a fact.

17 Renewable Reliability Score

The Renewable Reliability Score (RRS) is a proposed team-created analytical metric, not an official CEA metric. It is an internal, configurable summary designed for exploration, not compliance, certification, or dispatch.

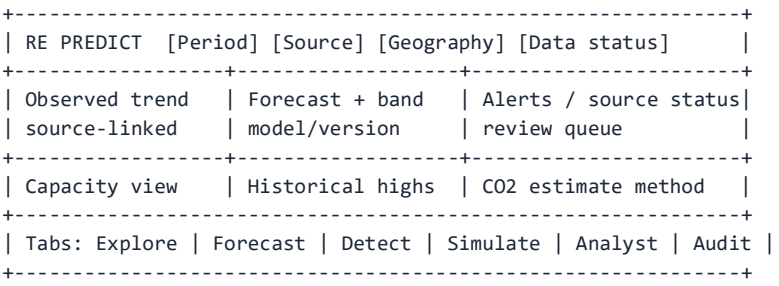
```
RRS = 100 x [w1 x Availability + w2 x ForecastAccuracy + w3 x Stability + w4 x DataCompleteness]
```

Each component is normalized to 0-1, weights sum to 1, and unavailable components are visibly handled rather than silently imputed. Availability requires an approved definition and source; ForecastAccuracy is based on held-out error relative to a baseline; Stability uses variability within an explicit rolling window; DataCompleteness measures valid expected observations. The score must show its component values, configuration, coverage and limitations.

18 Explainable AI and Natural Language Analyst

The analyst interface is retrieval-and-query grounded, not an open-ended answer generator. It translates questions into constrained semantic queries, retrieves approved observations and source evidence, performs deterministic calculations, and generates a cited narrative. Guardrails: answer only with retrieved evidence; show query filters and time range; label estimates; refuse ungrounded causal, policy, financial or plant-level claims; do not fabricate source fields; provide source document links; retain prompt, query, result IDs and model version in an audit log.

19 Dashboard and Product UX



Page	Exact widgets
Overview	Data freshness, observed daily generation cards, source-wise trend, installed-capacity view, source status.
Explore	Time series, period comparison, composition, regional view only where available, export with provenance.
Forecast	Target definition, horizon selector, point/interval chart, baseline comparison, model card and evaluation.
Detect	Alert list, severity, residual chart, evidence links, analyst disposition.
Simulate	Assumption editor, formula, low/base/high output chart, constraint messages, export scenario record.
Analyst	Question box, interpreted filters, cited answer, calculation steps, query audit.
Audit	Document catalog, parser status, data checks, transformation/model versions.

20 Backend API Architecture and Example Contracts

A REST API serves versioned read paths while asynchronous jobs handle ingestion, extraction, training and scenarios. Authentication scopes separate viewer, analyst, reviewer and administrator actions. All response objects include data_version and provenance references.

Endpoint	Purpose	Example contract
GET /v1/observations	Filtered approved observed values	?source_type=solar&frequency=daily&from=...; returns value, unit, period, provenance_id
GET /v1/forecasts/{run_id}	Registered forecast run	Returns target, horizon, points, interval, model_version, evaluation_summary
GET /v1/alerts	Alert triage queue	Returns method, severity, status, evidence, no asserted cause
POST /v1/scenarios	Execute stored formula on explicit assumptions	Body contains data_version, assumptions, formula_version; returns scenario_id and outputs
POST /v1/analyst/query	Constrained natural-language request	Returns interpreted query, answer, citations, limitation notices
GET /v1/provenance/{id}	Trace a value to source	Returns document URL, checksum, locator, extraction/review history

21 Database Design

```
source_document 1---* extraction_evidence *---1 observation
source_document 1---* capacity_observation
canonical data snapshot 1---* forecast_run 1---* forecast_point
canonical data snapshot 1---* scenario_run
observation 1---* alert
user 1---* analyst_audit_event
```

Use a relational metadata store (for documents, observations, runs, permissions and audit events) plus object storage for raw artifacts and a warehouse/time-series optimized serving layer for analytical queries. Relationships preserve many observations per document and many evidence rows per observation, avoiding loss of traceability.

22 Tech Stack

Layer	Proposed technologies	Reason
Frontend	React or Next.js; TypeScript; charting library	Fast interactive dashboard and typed UI.
Backend	Python FastAPI or Node service	Versioned APIs and background job integration.
ML	Python, pandas/polars, scikit-learn, LightGBM/XGBoost, statsmodels	Reproducible tabular and time-series experimentation.
Data	PostgreSQL, object storage, optional DuckDB/warehouse	Metadata integrity plus analytical performance.
Orchestration	Prefect, Airflow or scheduled containers	Observable ingestion/training jobs.
Deployment	Containers, managed compute, CI/CD	Portable staged releases.
Monitoring	Structured logs, metrics, traces, data/model dashboards	Operational and ML visibility.

23 Security Governance Provenance Auditability and Responsible AI

Control	Design
Access	Role-based access; least privilege; separate administrator/reviewer scopes.
Supply chain	Allow-list official sources; checksum raw files; scan artifacts; pin parser/model dependencies.
Provenance	Immutable raw artifact, source URL, extraction locator, data version and model version on outputs.
Privacy	No personal data is needed for initial public reporting workflow; minimize logs and protect credentials.
Responsible AI	Human review for alerts; no automated grid-control action; uncertainty and limitations visible; fairness/privacy review for future user data.
Audit	Append-only event records for ingestion, overrides, exports, prompts, scenario assumptions, and deployments.

24 MLOps

Model registry stages are candidate, reviewed, production and retired. Each run records code version, data snapshot, feature schema, hyperparameters, evaluation windows, metrics and model card. Monitor input freshness/missingness, feature distribution drift, residual distribution, interval coverage, latency and alert volume. Concept drift is investigated when post-deployment error or calibration materially changes. Retraining is scheduled or gated by data/quality evidence; rollback restores a prior approved model and retains the incident record.

25 Testing Strategy

Test class	Coverage

Test class	Coverage
Unit	Parsers, unit conversion, formulas, feature builders, alert thresholds, access checks.
Data	Schema contracts, source checksum, duplicates, period continuity, transformation reconciliation.
ML	Time split, leakage tests, baseline comparison, determinism where feasible, interval coverage.
API	Auth, validation, pagination, version headers, error contracts, provenance retrieval.
UI	Filter behavior, accessibility, labels for observed/predicted/simulated, source-link rendering.
Load	Concurrent dashboard queries and queued ingestion job behavior.
Security	Dependency scans, authorization tests, secret handling, injection and upload controls.
Acceptance	Persona scripts and judge-demo rehearsal against known data snapshot.

26 MVP and Production Roadmap

Stage	Included	Deliberately deferred
48-hour MVP	Selected official report catalog; manual/assisted extraction; normalized daily series; trend dashboard; seasonal-naive plus one candidate; one residual alert; one scenario; provenance screen.	Full automated PDF coverage, broad geography, weather connector, LLM free-form analysis.
Pilot	Automated validation/review queue; walk-forward model evaluation; alerts; analyst query with citations; monitoring.	High-scale multi-tenant controls and production SLAs.
Production	Hardened ingestion, roles, registry, drift monitoring, scalable API, governed external connectors.	Any feature that requires new source licensing or operational authority.

27 48 Hour Hackathon Implementation Plan

Hours	Priority deliverable	Owner focus
0-3	Agree target definition, source list, data contract, demo acceptance criteria.	Product + data
3-8	Ingest selected reports, save provenance, profile actual fields.	Data engineering
8-14	Normalize reviewed series; data quality checks; seed dashboard.	Data + frontend
14-20	Baseline forecast and time split; record measured metrics only if calculated.	ML
20-26	Residual anomaly, scenario formula, provenance panel.	ML + backend
26-34	Forecast/detect/simulate UI and analyst scripted response.	Frontend + backend
34-40	Integration tests, demo dataset snapshot, fallback screenshots.	All
40-46	Polish narrative, rehearsal, judges FAQ, limitations.	Product + presenter
46-48	Freeze version, verify links, deploy/demo contingency.	All

28 Suggested Team Roles and Work Split

Role	Primary responsibility	Handoff
Product lead/presenter	Story, scope, demo, acceptance criteria	Signed user journey and demo script
Data engineer	Catalog, extraction, normalization, provenance	Approved dataset snapshot
ML engineer	Baseline, walk-forward evaluation, anomaly/interval logic	Model card and reproducible run
Full-stack engineer	UI, API, deploy, audit trail	Stable demo build
Designer/QA	UX, labels, test scripts, visual polish	Demo-ready checklist

29 Five to Seven Minute Judge Demo

Time	Sequence	What to say or show
0:00-0:40	Problem and positioning	CEA is source of record; RE-PREDICT adds intelligence, not replacement reporting.
0:40-1:20	Sense	Open source catalog, report link, checksum/provenance and data status.
1:20-2:10	Understand	Filter source/period; compare observed trend with transparent units.
2:10-3:00	Predict	Show forecast, interval, model/version, baseline and measured-backtest panel if available.
3:00-3:45	Detect	Open a residual alert; show it is a review signal, not a declared cause.
3:45-4:40	Simulate	Adjust capacity/growth assumption; show formula and low/base/high conditional result.
4:40-5:30	Explain	Ask a constrained analyst question; reveal filters, citations and limitation.
5:30-6:30	Close	Show architecture, roadmap and why trust/provenance differentiates the product.

Do not claim deployment at scale, real-world savings, model accuracy, official endorsement, or causal diagnoses unless independently measured and evidenced during the demo.

30 Judging Differentiation and Novelty

- A full intelligence loop: Sense -> Understand -> Predict -> Detect -> Simulate -> Explain, rather than a static dashboard.
- Provenance first: every value, derived output and model run is traceable to a versioned official source artifact.
- Responsible AI by design: uncertainty, baselines, human review, and simulation-versus-prediction separation.
- Hackathon realism: a narrowly demonstrable slice with an extensible architecture rather than a fabricated nationwide claim.

31 Likely Judge Questions and Factual Answers

Question	Answer
Where does the data come from?	The official CEA Renewable Energy Portal and its reports page; the report includes daily, monthly and quarterly renewable reporting and under-construction project reporting. [1][2]
Is this an official CEA product?	No. It is a proposed analytical layer that preserves CEA as the authority for its published data.
How accurate is the forecast?	No accuracy is claimed before backtesting. We use a time-based holdout and compare with seasonal-naive baseline using MAE/RMSE and interval coverage.
Does an anomaly identify the cause?	No. It flags unusual behavior relative to a documented baseline and routes it for review.
Are carbon values official?	No. They are transparent factor-based estimates only when a documented factor is selected and shown.
What makes it scalable?	Immutable raw files, canonical versions, asynchronous jobs, model registry, and stateless APIs make a staged scale-out path possible.

32 KPIs

Category	Measured KPI	Target KPI to measure
Product	Task completion, time to source evidence, analyst feedback	Reduction in time to locate/report a cited insight.
Model	Holdout MAE/RMSE/MASE, interval coverage, calibration	Improvement or non-inferiority versus baseline and calibrated intervals.
Data quality	Coverage, freshness, parser review rate, reconciliation error	Pre-set quality thresholds by source/frequency.
Adoption	Active users, saved views, alert review time	Pilot-defined engagement and review targets.
Climate/energy	Only factor-method outputs with inputs/version	No impact claim without approved methodology

Category	Measured KPI	Target KPI to measure
		and measurement.

33 Risks Limitations and Mitigation

Risk	Mitigation
Report format changes	Version parsers; contract tests; review queue; preserve raw artifact.
Sparse/inconsistent history	Start with available target only; disclose coverage; use baseline; defer unsuitable targets.
Forecast misuse	Intervals, horizon labels, model card, no dispatch/control decision automation.
Spurious anomalies	Robust thresholds, data-quality gating, human disposition, feedback loop.
Wrong assumptions in scenarios	Require visible assumption form, constraints, saved version and sensitivity range.
Hallucinated analyst answer	Constrained query layer, retrieved citations, deterministic calculations, refusal rules.
Source availability	Cache permissible artifacts, detect retrieval failures, show freshness status; do not scrape beyond policy.

34 Deployment and Scalability Roadmap

Stage 1 uses a scheduled single-region deployment with object storage, Postgres and a worker queue. Stage 2 separates ingestion, extraction, API and ML workers; adds warehouse partitions by period/source and cacheable read models. Stage 3 adds multi-region resilience where required, observability SLOs, disaster-recovery exercises, policy-controlled connectors and tenant isolation. Scale should follow measured load, data volume and governance needs rather than premature infrastructure complexity.

35 Future Extensions

Extension	Value and guardrail
Weather APIs	Potential predictive features; retain source/licence/version and availability-at-issue-time to prevent leakage.
State/region granularity	Add only as supported by available official or approved data; preserve geographic definition.
Capacity/project data	Join compatible published capacity/project reporting with clear date/effective-status semantics.
Storage, EV, grid	Future integrations require separate data agreements and operational governance.
Probabilistic forecasting	Quantile/conformal approaches with calibration reporting.
Satellite/weather data	Validate licensing, resolution, latency and bias before production use.

36 Commercialization and Adoption Pathways

Potential adoption paths include a research/public-insight portal, an analyst workbench for energy organizations, a white-label data-intelligence module, or a governed API for approved partners. Potential value differs by stakeholder: public teams may value traceable reporting analysis; developers may value aggregate market context; researchers may value reproducible datasets. These are potential use cases, not claims of contracts, savings, performance, or endorsement.

37 Final Conclusion

RE-PREDICT is credible when it does not overclaim. Its role is to make authoritative reporting more usable through a governed intelligence layer: preserve evidence, make observations understandable, forecast with measured uncertainty, identify deviations for review, simulate transparent assumptions, and explain outputs with citations. The hackathon MVP should demonstrate this chain end to end on a limited, source-linked dataset and leave a clear path to production-grade validation.

Appendix A Equations and Definitions

Item	Expression / definition
Forecast residual	r_t = y_t - yhat_t
MAE	(1/n) sum |y_t - yhat_t|
RMSE	sqrt((1/n) sum (y_t - yhat_t)^2)
sMAPE	(100/n) sum [2|y-yhat|/(|y|+|yhat|)] with zero safeguards
MASE	MAE divided by in-sample naive MAE
Scenario generation	G = C x CF x H, conditioned on explicit assumptions
Avoided emissions estimate	AE = G x EF_counterfactual; optional net uses EF_counterfactual - EF_RE
RRS	100 x weighted sum of normalized availability, accuracy, stability, completeness

Appendix B API Example

```
GET /v1/observations?source_type=solar&frequency=daily&from=2026-09-01&to=2026-09-30
200 { "data_version":"dv_2026_10_02", "items":[{
  "period_start":"...", "value": "<observed>", "unit":"<source unit>",
  "provenance_id":"prov_...", "quality_status":"approved"
}] }
```

Appendix C Model Experiment Matrix

Experiment	Features	Validation	Decision rule
Seasonal naive	Prior seasonal/lag reference	Walk-forward	Required benchmark.
ETS/ARIMA	Target history and seasonal terms	Walk-forward	Keep if stable and competitive.
LightGBM/XGBoost	Allowed lags, rolling/calendar/capacity features	Walk-forward	Keep if improves primary metric and remains explainable.
LSTM/TFT later	Broad multi-series/history plus governed covariates	Walk-forward	Only if value exceeds complexity/latency cost.

Appendix D Demo Checklist

- Official source link opens; selected snapshot and checksum visible.
- Every observed chart has unit, period, filter and provenance path.
- Forecast shows baseline, horizon, model version and uncertainty label.
- Alert shows limitation: signal, not causal diagnosis.
- Scenario displays assumptions, formula version and simulation label.
- Analyst answer displays filters, citations and refusal behavior.

Appendix E Submission Checklist

- Problem, solution, architecture, demo and limitations are aligned.
- No unsupported model score, financial, emissions or impact claim.
- Source URLs and references retained; no fabricated data schema implied.
- Repository includes setup, data-provenance, model and demo notes.

Appendix F Glossary

Term	Meaning
Authoritative source	Publisher responsible for the official reported data.
Provenance	Evidence linking an output back to source artifact and transformation history.

Term	Meaning
Backtest	Evaluation using prior periods as if forecasting from that earlier point.
Prediction interval	Range intended to contain future observed value at a stated coverage level.
Data drift	Change in input distribution; concept drift is change in relationship between inputs and outcome.
Simulation	Conditional calculation under assumptions; distinct from prediction.

Appendix G References

[1] Central Electricity Authority, Government of India. Renewable Energy Portal - Reports. https://gen-re.cea.gov.in/reports (accessed 2 October 2026). The page lists daily renewable-generation reports, including entries through 30 September 2026, and provides monthly and quarterly report sections.

[2] Central Electricity Authority, Government of India. Renewable Energy Portal. https://gen-re.cea.gov.in/ (accessed 2 October 2026). Portal context for renewable-generation, installed-capacity, trend, historical-high and CO2-mitigation dashboard views, and source-wise reporting context.

Verified-fact note: The source reporting scope described in this report (daily, monthly and quarterly reports; solar, wind, hydro, geothermal, biomass and bagasse; and under-construction project reporting) is attributed to CEA portal/report material. Product architecture, metrics, models, formulas, targets and roadmaps are RE-PREDICT team proposals unless explicitly identified as observed or measured.